

Hamidreza Namjoomanesh

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Education

Islamic Azad University of Shiraz

Bachelor of Computer Engineering

Shiraz, Iran

Sep 2022 - Expected Jun 2026

- Cumulative GPA (up to 6th semester): 17.53 / 20 (3.6 / 4.0)
- Notable Courses: Data Structures, Algorithm Design, Artificial Intelligence, Compiler Design, Probability and Statistics for Engineers

Research Interests

• Natural Language Processing • Machine Learning • Deep Learning • Human-Computer Interaction

Research Experience

Research Assistant

Oct 2023 – Dec 2023

Islamic Azad University of Shiraz

Shiraz, Iran

- Contributed to an interdisciplinary research project focused on AI-aided design and modeling of engineering components using 3D printing.
- Developed a mobile application to present the background and technical aspects of the project to visitors and judges at the Iran 2023 Research and Technology Exhibition.
- Project was recognized among the top innovations nationwide. Earned a Research Assistant Fellowship and tuition discount for academic excellence.

Selected Projects

Advanced Code Plagiarism Detection Tool

Aug 2025

AI/Code Analysis/NLP-Inspired Project

- Developed a robust plagiarism detection system for Python code using **Abstract Syntax Trees (AST)**, **TF-IDF**, and multi-method similarity scoring (Jaccard, Cosine, Sequence Matching).
- Implemented **smart normalization** to detect copying even after variable renaming, function reordering, or whitespace manipulation.
- Designed an **NLP-inspired pipeline** treating code as structured language — leveraging techniques like vectorization and semantic fingerprinting.
- Generated interactive **network graphs** (using **networkx**) and comprehensive **statistical dashboards** (with **matplotlib & seaborn**) for visual insight into cheating patterns.
- Produced a risk-scored report with CSV export, enabling educators to identify high-risk cases efficiently.
- Available on GitHub: github.com/hrnrxb/Advanced-Code-Plagiarism-Detection-Tool

CourseAdvisor: RAG Chatbot for Academic Course Selection

Mar 2025

RAG Chatbot

- Available on Github Repo: github.com/hrnrxb/IAUCourseExp-Adviser-bot
- Developed a chatbot for querying real-world course & professor reviews using RAG (Retrieval-Augmented Generation). Combined FAISS, Pandas, and HF models for embedding to build a scalable question-answering system over semi-structured data.

Sarcasm-Aware Sentiment Classifier (RoBERTa + BiLSTM + Attention)

Jul 2025

NLP Classification Project

- Available on Github Repo: github.com/hrnrxb/Advanced-Sentiment-Classifer-RoBERTa-BiLSTM-Attention
- Built a hybrid deep model to classify sentiment with sarcasm & contrast-aware handling. Combined RoBERTa embeddings with BiLSTM + attention to improve detection on ambiguous and ironic reviews.
- Deployed interactive demo on HuggingFace Spaces using Gradio; trained on IMDB with custom sarcasm samples and error analysis.

AI News Aggregator Bot (Automated, Real-time)

Jul 2025

AI/ML/DL/NLP News Aggregation & Automation Project

- Available on GitHub: github.com/hrnrxb/AI-News-Aggregator-Bot
- Developed a robust Python-powered bot for real-time aggregation of cutting-edge AI, ML, DL, and NLP news from over 25 diverse sources (top research blogs, GitHub Trending, Hacker News, Reddit).
- Engineered a comprehensive system utilizing **RSS feeds**, **web scraping**, and **API integration** for efficient data collection.
- Implemented **persistent storage (SQLite)** for intelligent duplicate prevention across runs, ensuring unique news delivery.

- Deployed via **GitHub Actions** for fully automated, scheduled updates every 5 hours, demonstrating CI/CD best practices and secure credential handling.

Text-to-Speech (TTS) Bot with Adjustable Voice Parameters

Jul 2025

AI/NLP/Speech Synthesis Project

- Developed an interactive Text-to-Speech (TTS) bot for web deployment, enabling users to convert text to speech with adjustable parameters (Speed, Pitch, Volume).
- Engineered the system using **Coqui TTS** for high-quality audio generation and **Gradio** for a user-friendly interface.
- Implemented real-time waveform visualization for generated speech, enhancing user interaction and data presentation.
- Deployed successfully on **Hugging Face Spaces**, demonstrating practical proficiency in audio processing, web integration, and cloud deployment of AI models.
- Available on GitHub: github.com/hnrxb/TTS-with-Adjustable-Voice-Parameters

Teaching Experience

Chief Teaching Assistant

Apr 2024 – Now

Islamic Azad University of Shiraz

Shiraz, Iran

- Data Structures (for 3 consecutive semesters), Physics II Lab, Operation System Lab, Software Engineering Lab, Computer Networking Lab, Database Lab, Internet Engineering

Teaching Assistant

Sep 2022 – Jun 2023

Islamic Azad University of Shiraz

Shiraz, Iran

- Fundamental of Computer and Programming, Advanced Programming

Work Experience

Junior Data Analyst (Freelance)

Oct 2024 – Present

Remote

Hong Kong

- Led end-to-end data analysis projects for international clients across various industries, working remotely from Iran.
- Specialized in Natural Language Processing (NLP), performing unstructured text mining, trend detection, and insight extraction using Python and Pandas.
- Built and evaluated NLP pipelines for document classification, keyword extraction, and semantic search.
- Collaborated on several computer vision tasks, gaining experience in image processing and feature extraction using OpenCV and PyTorch.

Founder, Artificial Intelligence Association of Shiraz IAU

Nov 2024 – Sep 2025

Islamic Azad University of Shiraz

Shiraz, Iran

- Founded and lead the First official AI association at Islamic Azad University of Shiraz, organizing academic events, technical workshops, and collaboration spaces for AI enthusiasts.
- Organized academic events, technical workshops, and interdisciplinary meetups focused on AI, ML, and NLP.

Technical Skills

Programming Languages: Python, C/C++, SQL, HTML/CSS, JavaScript (basic)

Machine Learning & Deep Learning: PyTorch, Scikit-Learn, Keras, XGBoost, LightGBM

Natural Language Processing: HuggingFace Transformers, spaCy, NLTK, Gensim, Pattern, TextBlob, CoreNLP, Polyglot, Fairseq, Jina

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Experiment Tracking & Deployment: Weights & Biases (WandB), Gradio, Streamlit, Docker (basic)

Databases & Tools: MySQL, MongoDB, FAISS, LangChain

Version Control & Collaboration: Git, GitHub

Soft Skills: Critical Thinking, Problem-Solving, Technical Writing, Presentation, Team Collaboration

Languages: Persian (Native), English (Professional)

Honors & Awards

Research Assistant Fellowship Award

December 2023

Islamic Azad University of Shiraz

- Awarded RA Fellowship and tuition discount in recognition of academic excellence and research impact during national research exhibition.

Top 5% of Entering Class (2022)

Ongoing

Islamic Azad University of Shiraz

- Ranked among top 5% of students in the 2022 entering cohort.

References

Dr. Amin Eskandari

Assistant Professor

- Computer Engineering Department, Islamic Azad University of Shiraz, Shiraz, Iran
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Dr. Asghar Jahed

Assistant Professor

- Computer Engineering Department, Islamic Azad University of Shiraz, Shiraz, Iran
- E-mail: jahed20231403@gmail.com